TISSUE REPAIR & RECOVERY

TB-500 — the full review

SYNTHETIC FRAGMENT OF THYMOSIN BETA-4

This is the deeper look behind the short version — why the related-protein research doesn't carry over to TB-500 itself, what the one direct test of the molecule actually found, and what your consultation covers.

What it is

TB-500 is a lab-made chain of seven amino acids, built to copy one small stretch — positions 17 through 23 — of a much larger, naturally occurring 43-amino-acid protein called thymosin beta-4, a protein the body already makes in nearly every cell type, especially at sites of injury and inflammation. TB-500 carries an added acetyl chemical group on the front end of the chain, a modification FDA's own reviewers state "irreversibly alters" the molecule's charge, size, and behavior. TB-500 is not simply a piece of thymosin beta-4 — it is a chemically modified piece, and its activity in the body cannot be assumed to match either the unmodified natural fragment or the full parent protein.

TB-500 is sold almost exclusively as a self-administered subcutaneous or intramuscular injection, most often through online "research use only" vendors, and is most often used for tendon, ligament, and muscle injury recovery, general wound healing, and vascular support. It is most often used for those recovery goals, and the evidence commonly cited for that use comes mainly from studies of the related, unmodified natural protein rather than from TB-500 itself.

TB-500 should not be confused with two other "thymosin"-family names that circulate in wellness marketing. Thymosin alpha-1 is a structurally unrelated peptide, approved in more than 30 countries for hepatitis and as cancer-treatment support (though not in the US), that works through a different immune-activating mechanism. Thymalin is a different substance again. The shared "thymosin" name reflects a shared origin in 1960s–70s thymus-tissue research, not a shared mechanism or safety record.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Tendon, ligament, and muscle injury recovery	**Animal data only**	No human study of any kind exists. What animal evidence there is comes almost entirely from the full-length protein or its unmodified fragment, not TB-500 itself.
General wound and skin healing	**Animal data only**	Supporting studies used a related, unmodified protein in aged mice. When TB-500 itself was directly tested in a 2024 in vitro assay, it did not induce wound healing at the tested concentration.
Cardiac tissue repair after heart attack	**Limited human data**	A published study associated naturally elevated blood levels of the parent protein with improved outcomes after stem-cell therapy — a correlational biomarker finding, not a trial of administered TB-500.

USED FOR	EVIDENCE	WHAT THAT MEANS
Hair growth	**Animal data only**	Mouse studies found the full-length parent protein activates hair follicle stem cells and accelerates hair growth. No study, animal or human, has tested TB-500 specifically for this use.
Vascular support and general recovery	**Theoretical**	A plausible mechanism from related-protein research, with no direct study of TB-500 for this use, human or animal.
Post-surgical or preventive use in healthy people	**Not studied**	No study of any kind has evaluated TB-500 for prophylactic or general-wellness use in people without an existing injury. This use exists in marketing only.

How it works

The proposed mechanism starts from real, well-established cell biology: a six-amino-acid core within TB-500's sequence is known as the "actin-binding region" of thymosin beta-4, and the parent protein does bind actin as part of normal cell function. Actin filaments are the internal scaffolding cells use to change shape and move, so a molecule that influences actin dynamics is mechanistically plausible as an influence on the cell migration involved in wound repair. Marketing extends that story further, to new blood vessel formation and general tissue repair.

The complication is that essentially none of the underlying research demonstrating actin-binding, cell migration, or wound-healing effects was performed on TB-500 itself. The most frequently cited supporting study tested topical full-length thymosin beta-4 and the plain, non-acetylated fragment in aged mice — not TB-500. FDA's reviewers are explicit that "the pharmacological profile of the non-acetylated heptapeptide LKKTETQ cannot be directly extrapolated to the N-acetylated heptapeptide TB-500."

When TB-500 itself was actually tested — a 2024 study using confluent fibroblast cultures — it did not induce wound healing at the tested concentration. One of TB-500's own breakdown products did show wound-healing activity in the same experiment, an unresolved finding suggesting the molecule sold and injected as "TB-500" may not be the active agent at all, and that whatever it breaks down into inside the body has not itself been fully characterized.

What the research actually looked at

This is where TB-500 diverges most sharply from other compounds in this catalog: the honest number of documented human subjects is zero. FDA's July 2026 review states plainly that "the nomination did not include, and FDA did not find any information in the medical literature where, TB-500 was administered to patients to treat any disease or condition" — no human has ever been studied taking TB-500, by any route, for any indication, published anywhere. By comparison, BPC-157 — a related but distinct compound in this same catalog — has roughly 81 documented human subjects across its published literature; TB-500's record is meaningfully thinner still.

No formal toxicology of any kind exists either: no acute, repeat-dose, genotoxicity, reproductive, or carcinogenicity studies have been conducted in any species. Animal pharmacokinetic data is also thin and off-route — the only data identified comes from horses given a subcutaneous dose, not the intramuscular route TB-500 is marketed for, with blood levels so low they were undetectable within hours.

A 2026 scoping review that mapped 80 studies across the entire thymosin beta-4 and TB-500 literature found that only one study tested TB-500 itself directly — the same 2024 in vitro/rat metabolite experiment described above — and identified "no human interventional studies of

administered TB4 or TB-500" in tendon, ligament, muscle, bone, cartilage, fat-graft, or spine categories, which covers essentially every musculoskeletal use TB-500 is actually marketed for.

Common misunderstandings

- **“TB-500 is the same thing as thymosin beta-4.”** It is not. TB-500 is a synthetic, chemically modified 7-amino-acid fragment of the 43-amino-acid parent protein, and FDA's own reviewers state the pharmacology of the related, unmodified fragment "cannot be directly extrapolated to" TB-500.
- **“There's a large, credible research base behind TB-500.”** There is a large research base behind the full-length parent protein. There is essentially no research base behind TB-500 the fragment — a comprehensive 2026 literature mapping found exactly one study, out of 80 in the broader field, that tested TB-500 directly.
- **“The FDA is about to approve TB-500 for compounding.”** FDA's own scientists formally recommended against adding TB-500 to the compounding bulks list, citing zero efficacy evidence and zero human safety data. An advisory committee then voted 8-6, with one abstention, to recommend it anyway — a non-binding vote FDA has not adopted. TB-500 remains off the compounding bulks list.
- **“It's part of the same 'thymosin' family as approved drugs, so it's basically proven safe.”** Thymosin alpha-1 and thymosin beta-4/TB-500 are structurally and functionally unrelated peptides that happen to share a name from a shared 1960s–70s thymus-tissue origin, not a shared mechanism or safety record.

Who it is not for

- Active cancer, a cancer history, an undiagnosed mass, or overdue cancer screening — no carcinogenicity studies of TB-500 have ever been performed, in any species.
- Autoimmune disease or a history of drug hypersensitivity reactions — FDA specifically flags immunogenicity risk for injectable peptides like this one.
- Pregnancy, breastfeeding, or trying to conceive — no reproductive or developmental toxicity study has been conducted in any species.
- Anyone relying on it in place of a proven treatment — physical therapy, an approved medication, image-guided injection, or surgery — for a condition with an actual evidence-based option.
- Anyone sourcing it outside a licensed, regulated pathway — TB-500 is not currently on the 503A Bulks List, so no legal compounding pathway exists for it today.
- Competitive athletes subject to WADA-code testing.

Competitive athletes: TB-500 is explicitly named and prohibited at all times, in and out of competition, under WADA category S2.3, which lists "Thymosin-β4 and its derivatives e.g. TB-500." Anti-doping testing operates on strict liability, and WADA has funded detection methods targeting TB-500's metabolites since 2013, so a clinician's recommendation or a compounding label offers no protection once it is detected.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and there is no human exposure data at all. FDA's own July 2026 review found no record of TB-500 ever being given to a documented human being. An advisory committee voted 8-6, with one abstention, to recommend it for wound healing; advisory only, not adopted.

What's sold is a chemically modified fragment, not the full natural protein. The acetylated 7-amino-acid piece is distinct enough from full-length thymosin beta-4 that FDA states findings on one cannot be assumed to carry over to the other.

Everything claimed rests on animal research of a related, but different, molecule.

Clinical detail

TB-500 is the acetylated 7-amino-acid fragment (Ac-LKKTETQ) copying positions 17–23 of the 43-amino-acid parent protein thymosin beta-4.

FDA's July 2026 review found no record of TB-500 ever being administered to a documented human being, by any route, for any condition.

A July 23–24, 2026 Pharmacy Compounding Advisory Committee voted 8-6, with one abstention, to recommend TB-500 for the 503A Bulks List, overruling FDA staff's recommendation against inclusion; the vote is advisory only and has not been adopted.

No confirmed FDA warning letter names TB-500 specifically, unlike several other compounds in this catalog.

A structurally different, full-length version of the parent protein is in real FDA-regulated drug development under three separate programs, none yet approved for any use.

> The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.
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> TB-500 is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.
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> Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.
>
> Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.